

RStudio

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Go to file/function Addins

Project: (None)

```
1 lower <- c(1, 5, 15, 20, 50, 80)
2 upper <- c(5, 15, 20, 50, 80, 110)
3 freq <- c(200, 450, 300, 1500, 700, 44)
4 N <- sum(freq)
5 cf <- cumsum(freq)
6 median_position <- N / 2
7 median_class <- which(cf >= median_position)[1]
8 L <- lower[median_class]
9 f <- freq[median_class]
10 CF <- ifelse(median_class == 1, 0, cf[median_class - 1])
11 h <- upper[median_class] - lower[median_class]
12 median_value <- L + ((median_position - CF) / f) * h
13 cat("Total frequency =", N, "\n")
14 cat("Median position =", median_position, "\n")
15 cat("Median class =", lower[median_class], "-", upper[median_class], "\n")
16 cat("Approximate Median =", median_value, "\n")
17
18 |
```

Environment History Connections Tutorial

R - Global Environment

group	chr	[1:12]	"A" "A" "A" "A" "A" "A" "A" "B" "B" "B" "B" "B"
h		30	
L		20	
lower	num	[1:6]	1 5 15 20 50 80
max_frequency		4L	
mean_age		29.962962962963	
median_age		max_frequency (integer, 56 bytes)	
median_class		4L	
median_position		1597	
median_value		32.94	

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Zoom Export

Console Terminal Background Jobs

R - R4.5.1 - ~/

```
> h <- upper[median_class] - lower[median_class]
> median_value <- L + ((median_position - CF) / f) * h
> cat("Total frequency =", N, "\n")
Total frequency = 3194
> cat("Median position =", median_position, "\n")
Median position = 1597
> cat("Median class =", lower[median_class], "-", upper[median_class], "\n")
Median class = 20 - 50
> cat("Approximate Median =", median_value, "\n")
Approximate Median = 32.94
> |
```

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RStudio

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Go to file/function Addins

median.r 2meanmedian.r

```
1 age <- c(13, 15, 16, 16, 19, 20, 20, 21, 22, 22,
2         25, 25, 25, 25, 30, 33, 33, 35, 35, 35,
3         35, 36, 40, 45, 46, 52, 70)
4 mean_age <- mean(age)
5 median_age <- median(age)
6 frequency <- table(age)
7 max_frequency <- max(frequency)
8 modes <- names(frequency[frequency == max_frequency])
9 cat("Mean =", mean_age, "\n")
10 cat("Median =", median_age, "\n")
11 cat("Mode =", paste(modes, collapse = ", "), "\n")
12 cat("Modality = Bimodal\n")
```

Run Source

Environment History Connections Tutorial

R Global Environment

lower	num	[1:6] 1 5 15 20 50 80
max_frequency	4L	
mean_age	29.962962962963	
median_age	25	
median_class	4L	
median_position	1597	
median_value	32.94	
modes	chr	[1:2] "25" "35"
N	3194	
upper	num	[1:6] 5 15 20 50 80 110

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9:1 [Top Level] R Script

Console Terminal Background Jobs

R 4.5.1 ~ /

```
> # Mode
> modes <- names(frequency[frequency == max_frequency])
> cat("Mean =", mean_age, "\n")
Mean = 29.96296
> cat("Median =", median_age, "\n")
Median = 25
> cat("Mode =", paste(modes, collapse = ", "), "\n")
Mode = 25, 35
> cat("Modality = Bimodal\n")
Modality = Bimodal
>
```

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RStudio

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Go to file/function Addins

Project: (None)

Environment History Connections Tutorial

Import Dataset 139 MiB

Global Environment

median_value	32.94
min_max	num [1:5] 0 0.125 0.25 0.5 1
min_value	200
modes	chr [1:2] "25" "35"
N	3194
sd_value	282.842712474619
upper	num [1:6] 5 15 20 50 80 110
values	num [1:12] 7 8 5 6 12 14 8 10 9 15 ...
x	num [1:12] 7 8 5 6 12 14 7 10 9 8 ...
z_score	num [1:5] -1.061 -0.707 -0.354 0.354 1.768

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Zoom Export

```
1 data <- c(200, 300, 400, 600, 1000)
2 min_value <- min(data)
3 max_value <- max(data)
4 min_max <- (data - min_value) / (max_value - min_value)
5 mean_value <- mean(data)
6 sd_value <- sqrt(mean((data - mean_value)^2))
7 z_score <- (data - mean_value) / sd_value
8 cat("Original Data:", data, "\n")
9 cat("Min-Max Normalization:", round(min_max, 3), "\n")
10 cat("Mean =", mean_value, "\n")
11 cat("Standard Deviation =", round(sd_value, 3), "\n")
12 cat("Z-Score Normalization:", round(z_score, 3), "\n")
```

12:55 [Top Level] R Script

Console Terminal Background Jobs

R 4.5.1 ~ /

```
> cat("Original Data:", data, "\n")
Original Data: 200 300 400 600 1000

> cat("Min-Max Normalization:", round(min_max, 3), "\n")
Min-Max Normalization: 0 0.125 0.25 0.5 1

> cat("Mean =", mean_value, "\n")
Mean = 500

> cat("Standard Deviation =", round(sd_value, 3), "\n")
Standard Deviation = 282.843

> cat("Z-Score Normalization:", round(z_score, 3), "\n")
Z-Score Normalization: -1.061 -0.707 -0.354 0.354 1.768
>
```

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Go to file/function Addins

Project: (None)

```
11 bin_median <- lapply(bins, function(x) rep(median(x), length(x)))
12
13 # (c) Smoothing by Bin Boundaries
14 bin_boundaries <- lapply(bins, function(x) {
15   low <- min(x)
16   high <- max(x)
17   sapply(x, function(v) {
18     if (abs(v - low) <= abs(v - high)) low else high
19   })
20 })
21
22 cat("\nOriginal Bins:\n")
23 print(bins)
24
25 cat("\nSmoothing by Bin Mean:\n")
26 print(bin_mean)
27
28 cat("\nSmoothing by Bin Median:\n")
29 print(bin_median)
30
31 cat("\nSmoothing by Bin Boundaries:\n")
32 print(bin_boundaries)
```

Environment History Connections Tutorial

R - Global Environment

data	num [1:24] 11 13 13 15 15 16 19 20 20 20 ...
f	1500
freq	num [1:6] 200 450 300 1500 700 44
frequency	'table' int [1:17(1d)] 1 1 2 1 2 1 2 4 1 2 ...
group	chr [1:12] "A" "A" "A" "A" "A" "A" "B" "B" "B" "B" "B"
h	30
L	20
lower	num [1:6] 1 5 15 20 50 80
max_frequency	4L
max_value	1000

Files Plots Packages Help Viewer Presentation

Console Terminal Background Jobs

R - R4.5.1 - ~/

```
> print(bin_boundaries)
$`1`
[1] 11 11 11 11 16 16

$`2`
[1] 19 19 19 19 21 21

$`3`
[1] 22 22 22 22 45 45

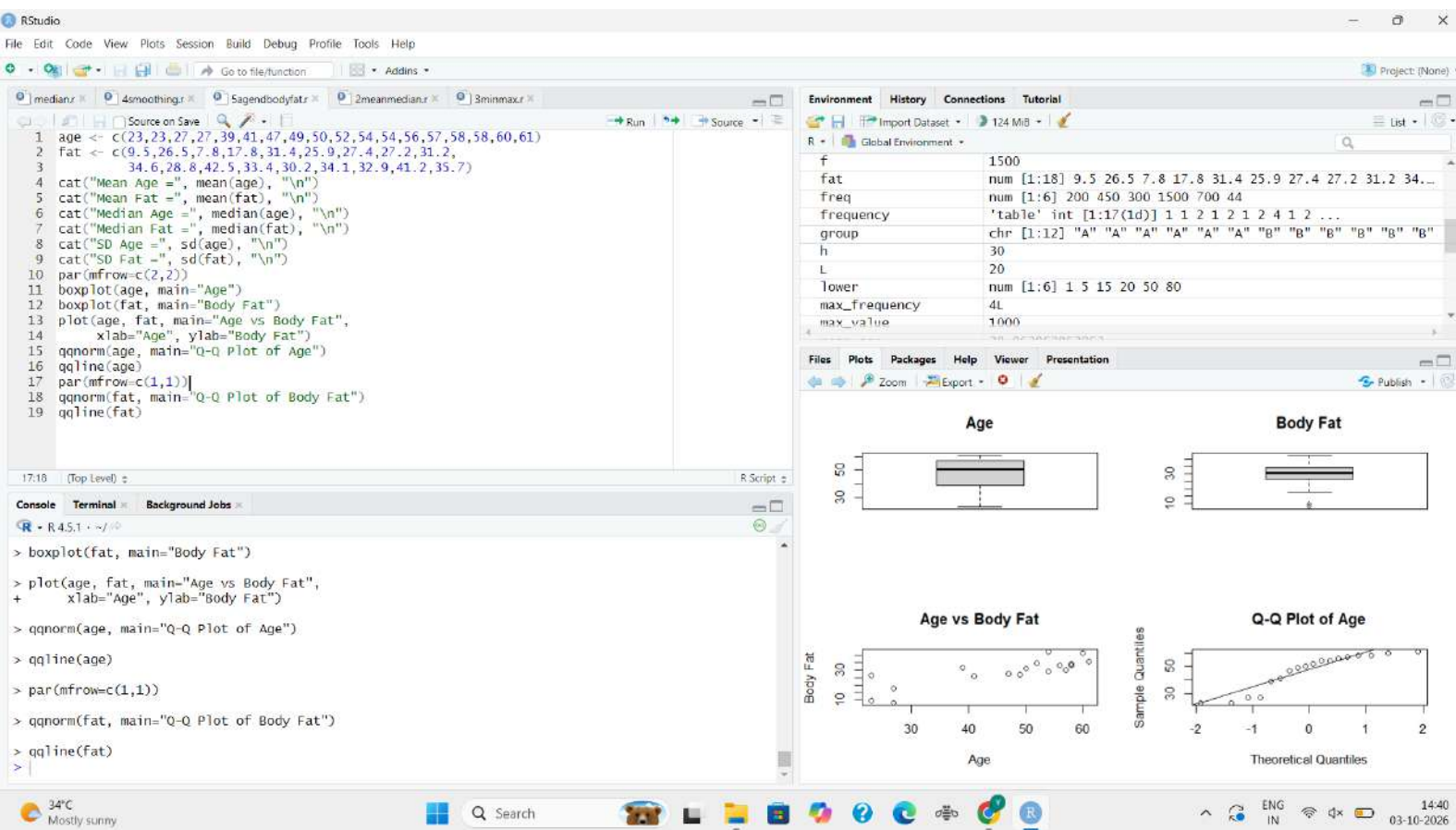
$`4`
[1] 45 45 75 75 75 75

>
```

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Go to file/function Addins

medians.r 4smoothings.r Sagendbodyfat.r 6mmrads.r 2meanmedian.r 3minmax.r

Run Source

```
5 x <- 35
6
7 # (i) Min-Max Normalization
8 min_age <- min(age)
9 max_age <- max(age)
10
11 min_max <- (x - min_age) / (max_age - min_age)
12
13 # (ii) Z-Score Normalization
14 mean_age <- mean(age)
15 sd_age <- 12.94
16
17 z_score <- (x - mean_age) / sd_age
18
19 # (iii) Decimal Scaling
20 decimal <- x / 100
21
22 # Output
23 cat("Min-Max Normalization =", min_max, "\n")
24 cat("Z-Score Normalization =", z_score, "\n")
25 cat("Decimal Scaling =", decimal, "\n")
26
```

26:1 (Top Level)

R Script

Environment History Connections Tutorial

R - Global Environment

min_max	0.315789473684211
min_value	200
modes	chr [1:2] "25" "35"
N	3194
sd_age	12.94
sd_value	282.842712474619
upper	num [1:6] 5 15 20 50 80 110
values	num [1:12] 7 8 5 6 12 14 8 10 9 15 ...
x	35
z_score	-0.88442383651039

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Zoom Export

Console Terminal Background Jobs

R - R 4.5.1 - ~/

```
> # (iii) Decimal Scaling
> decimal <- x / 100
>
> # Output
> cat("Min-Max Normalization =", min_max, "\n")
Min-Max Normalization = 0.3157895
> cat("Z-Score Normalization =", z_score, "\n")
Z-Score Normalization = -0.8844238
> cat("Decimal Scaling =", decimal, "\n")
Decimal Scaling = 0.35
> cat("Decimal Scaling =", decimal, "\n")
Decimal Scaling = 0.35
>
```

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RStudio

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Go to file/function Addins

median.r 4smoothings.r Sagendbodyfat.r 6mmzds.r pencil7.r 2meanmedian.r 3minmax.r

Source on Save Run Source

```
1 # Create vector
2 pencils <- c(9, 25, 23, 12, 11, 6, 7, 8, 9, 10)
3
4 # Mean
5 mean_value <- mean(pencils)
6
7 # Median
8 median_value <- median(pencils)
9
10 # Mode
11 frequency <- table(pencils)
12 mode_value <- names(frequency)[frequency == max(frequency)]
13
14 # Output
15 cat("Mean =", mean_value, "\n")
16 cat("Median =", median_value, "\n")
17 cat("Mode =", mode_value, "\n")
```

17:32 [Top Level] R Script

Console Terminal Background Jobs

R • R 4.5.1 • ~/

```
> # Mode
> frequency <- table(pencils)
> mode_value <- names(frequency)[frequency == max(frequency)]
>
> # Output
> cat("Mean =", mean_value, "\n")
Mean = 12
> cat("Median =", median_value, "\n")
Median = 9.5
> cat("Mode =", mode_value, "\n")
Mode = 9
>
```

Environment History Connections Tutorial

R • Global Environment

mode_value	"9"
modes	chr [1:2] "25" "35"
N	3194
pencils	num [1:10] 9 25 23 12 11 6 7 8 9 10
sd_age	12.94
sd_value	282.842712474619
upper	num [1:6] 5 15 20 50 80 110
values	num [1:12] 7 8 5 6 12 14 8 10 9 15 ...
x	35
z_score	-0.88442383651039

Files Plots Packages Help Viewer Presentation

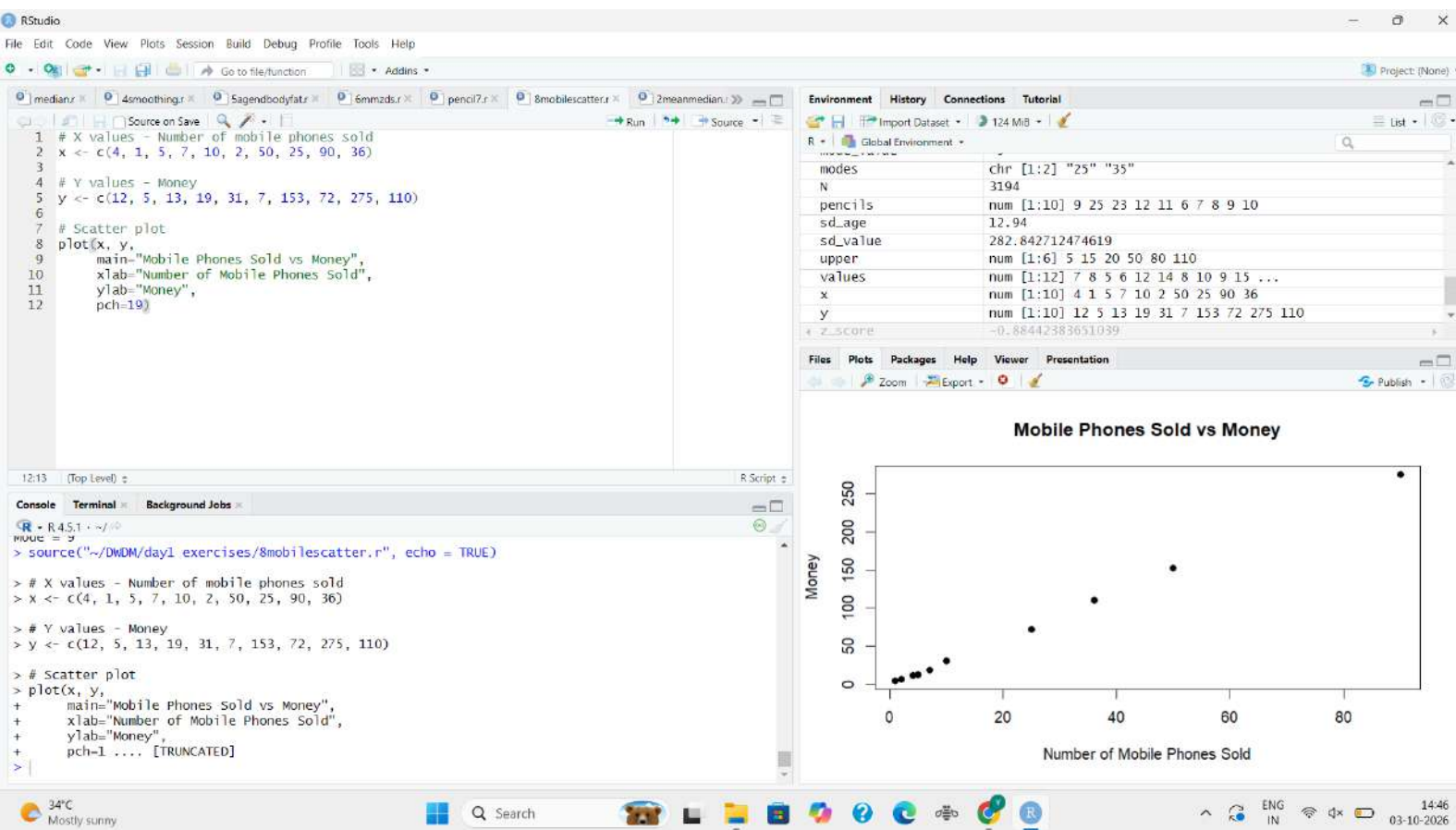
Zoom Export

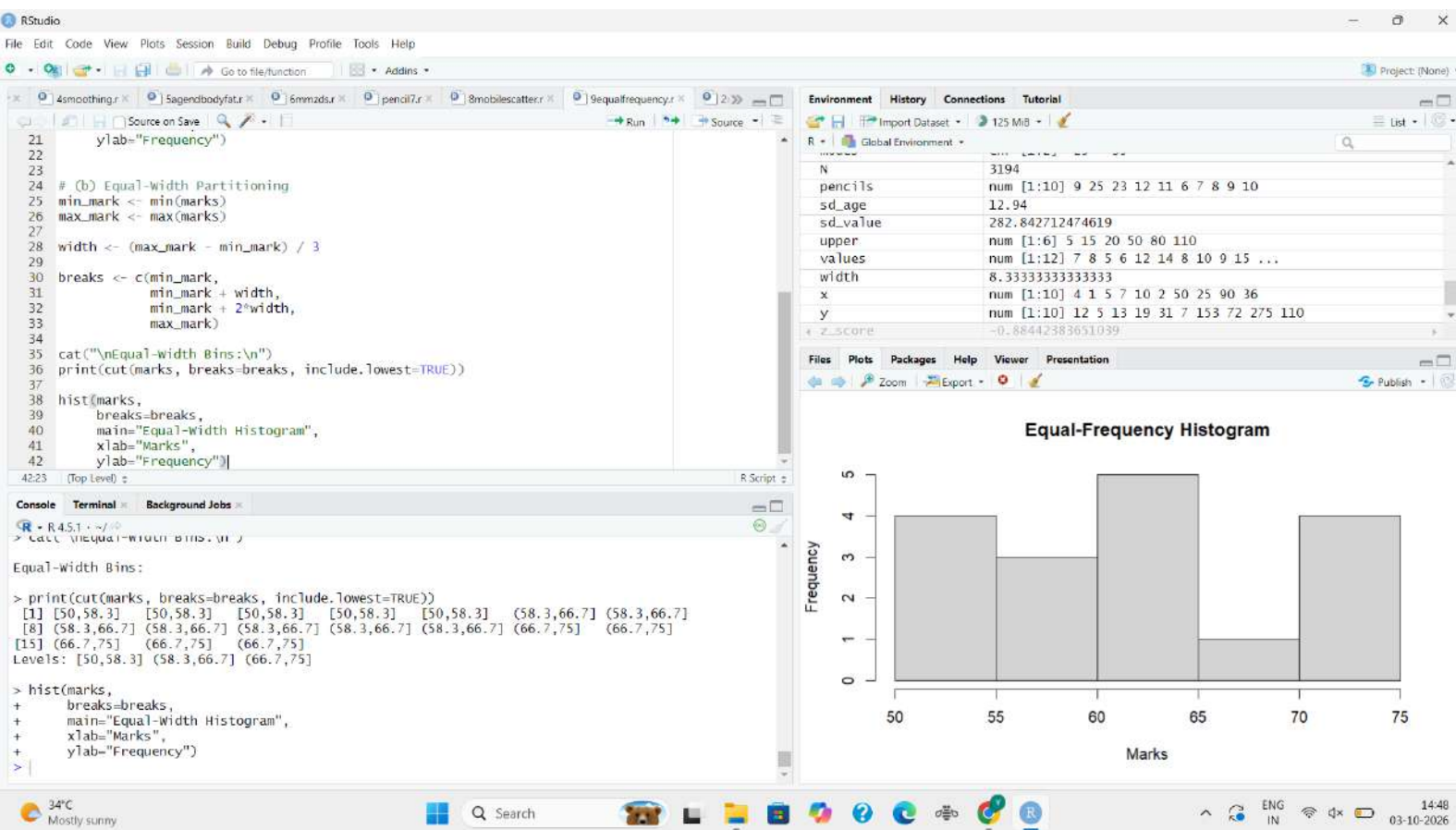
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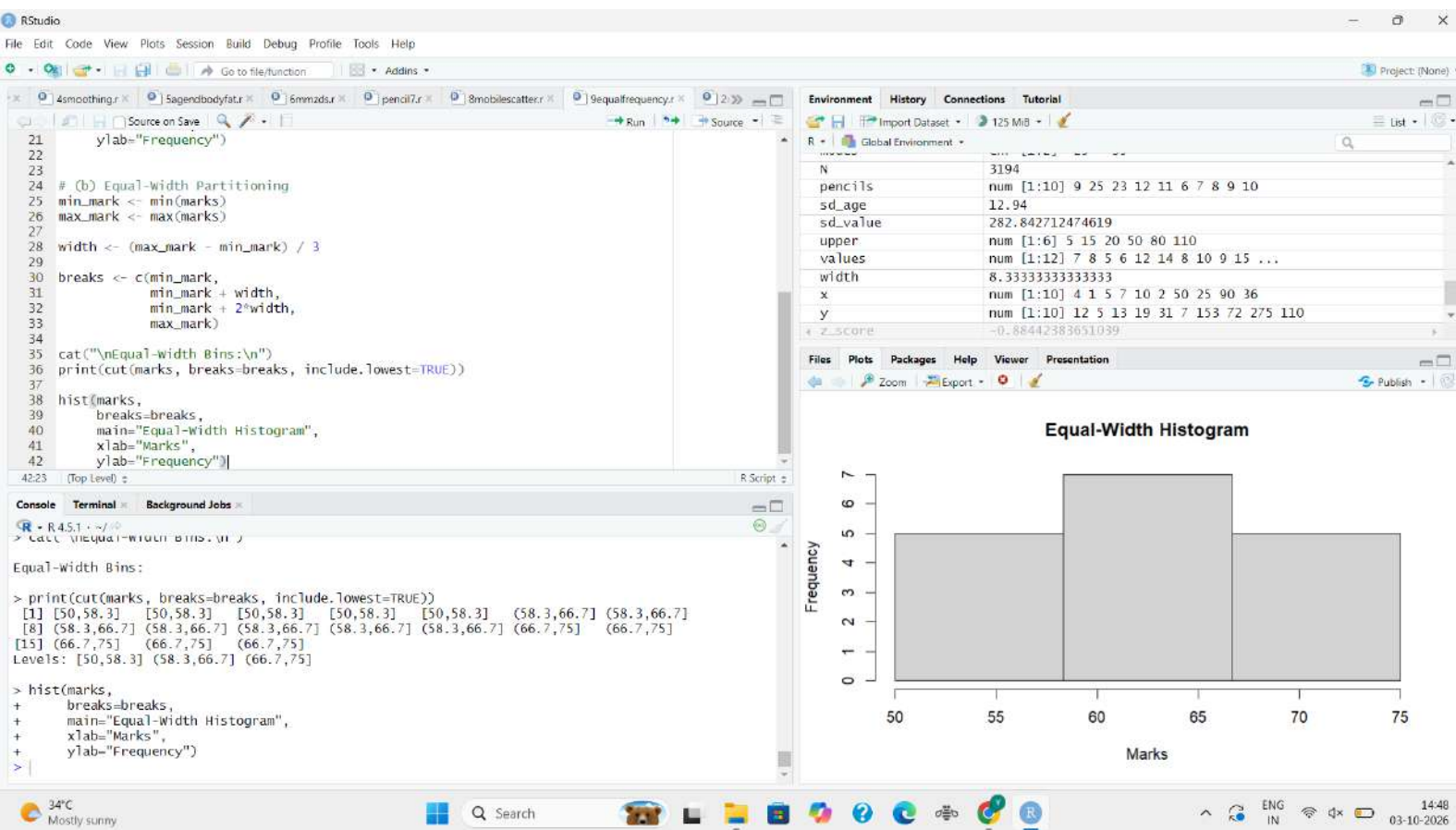
Search

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RStudio

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Go to file/function Addins

5agendbodyfat.r 6mmzds.r 7pendi7.r 8mobilesca.r 9equalfrequency.r 10rangends.r 11zmeanm.r

Source on Save

```
1 # Speed data
2 speed <- c(78.3, 81.8, 82, 74.2, 83.4,
3           84.5, 82.9, 77.5, 80.9, 70.6)
4
5 # Interquartile Range
6 IQR_value <- IQR(speed)
7
8 # Standard Deviation
9 SD_value <- sd(speed)
10
11 cat("Interquartile Range =", IQR_value, "\n")
12 cat("Standard Deviation =", SD_value, "\n")
```

Run Source

Environment History Connections Tutorial

R - Global Environment

sd_age	num	12.94
sd_value	num	282.842712474619
SD_value	num	4.44583450484208
speed	num	[1:10] 78.3 81.8 82 74.2 83.4 84.5 82.9 77.5 80.9 70.6
upper	num	[1:6] 5 15 20 50 80 110
values	num	[1:12] 7 8 5 6 12 14 8 10 9 15 ...
width	num	8.33333333333333
x	num	[1:10] 4 1 5 7 10 2 50 25 90 36
y	num	[1:10] 12 5 13 19 31 7 153 72 275 110
z_score	num	-0.88442383651039

Files Plots Packages Help Viewer Presentation

Zoom Export

12:44 [Top Level] R Script

Console Terminal Background Jobs

```
R - R45.1 - ~/...
> # speed data
> speed <- c(78.3, 81.8, 82, 74.2, 83.4,
+           84.5, 82.9, 77.5, 80.9, 70.6)
>
> # Interquartile Range
> IQR_value <- IQR(speed)
>
> # Standard Deviation
> SD_value <- sd(speed)
>
> cat("Interquartile Range =", IQR_value, "\n")
Interquartile Range = 4.975
>
> cat("Standard Deviation =", SD_value, "\n")
Standard Deviation = 4.445835
>
```

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Search

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RStudio

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Go to file/function Addins

6mmzdsr.R pendl7.R 8mobilescatter.R 9equalfrequency.R 10rangends.R 11qsfnd.R 2mean.R

```
1 # Age data
2 age <- c(13,15,16,16,19,20,20,21,22,22,
3         25,25,25,25,30,33,33,35,35,35,
4         35,36,40,45,46,52,70)
5
6 # Find Q1 and Q3
7 Q1 <- quantile(age, 0.25)
8 Q3 <- quantile(age, 0.75)
9
10 cat("First Quartile (Q1) =", Q1, "\n")
11 cat("Third Quartile (Q3) =", Q3, "\n")
```

Environment History Connections Tutorial

R - Global Environment

Values

age	num [1:27]	13 15 16 16 19 20 20 21 22 22 ...
bin1	num [1:6]	50 55 55 55 58 59
bin2	num [1:6]	60 61 63 63 65 65
bin3	num [1:5]	67 71 71 72 75
breaks	num [1:4]	50 58.3 66.7 75
cf	num [1:6]	200 650 950 2450 3150 ...
CF		950
data	num [1:24]	11 13 13 15 15 16 19 20 20 20 ...
decimal		0.35

Files Plots Packages Help Viewer Presentation

11:39 [Top Level] R Script

Console Terminal Background Jobs

```
R - R4.5.1 ~\...
> # Age data
> age <- c(13,15,16,16,19,20,20,21,22,22,
+         25,25,25,25,30,33,33,35,35,35,
+         35,36,40,45,46,52,70)
>
> # Find Q1 and Q3
> Q1 <- quantile(age, 0.25)
>
> Q3 <- quantile(age, 0.75)
>
> cat("First Quartile (Q1) =", Q1, "\n")
First Quartile (Q1) = 20.5
>
> cat("Third Quartile (Q3) =", Q3, "\n")
Third Quartile (Q3) = 35
>
```

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